Aerospace 311: Project 1

Assigned: 10/2/2026 **Due:10/16/2026**

The objective of this assignment is to determine the sectional lift on an airfoil using the flowfield around it. This will be done by calculating the circulation and then using the Kutta-Joukowski lift theorem to solve for the lift coefficient. Compare the resulting lift coefficient to the value obtained from a Computational Fluid Dynamics (CFD) solution. This is a homework project and requires a short write-up as part of the submission and can be completed and submitted with a partner. See the description below:

1. Background

The Kutta-Joukowski theorem is a fundamental theorem in theoretical aerodynamics. Written as,

$$L' = \rho_\infty U_\infty \Gamma_{aerodynamic} \tag{1}$$

it relates the aerodynamic circulation around an airfoil to the lift it generates. L' the lift per unit length, ρ_∞ and U_∞ are the freestream density and speed, and $\Gamma_{aerodynamic}$ is the circulation integrated along a closed-contour integration path, $\vec{s}$. $\Gamma_{aerodynamic}$ is calculated with,

$$\Gamma_{aerodynamic} = -\oint_S \vec{U} \cdot d\vec{s} \tag{2}$$

where $\vec{U}$ is the local velocity vector and $d\vec{s}$ is an infinitesimal element along $\vec{s}$ (using a counterclockwise positive path). The lift coefficient c_l can then be determined with the relation for sectional lift, $L' = 0.5\rho_\infty U_\infty^2 c c_l$ (with c being the airfoil chord length), such that,

$$c_l = \frac{2\Gamma_{aerodynamic}}{U_\infty c} \tag{3}$$

Recall that the relations for the Reynolds and Mach numbers are,

$$Re = \frac{\rho_\infty U_\infty c}{\mu_\infty} \tag{4}$$

$$M = \frac{U_\infty}{a_\infty} = \frac{U_\infty}{\sqrt{\gamma R T_\infty}} \tag{5}$$

where $R = 1716.5\ \frac{ft*lb_f}{slug*°R}$ and $\gamma = 1.4$. In CFD, flow conditions are often set to match a desired Reynolds number and Mach number, so these relations can help determine unknown dimensional flow quantities when a Reynolds and Mach are specified.

2. Task Description

Computational Fluid Dynamics (CFD) uses mathematical models to simulate flows around aerodynamic bodies. A CFD simulation has been performed on the E387 airfoil, in which the conditions are $Re = 460{,}000$, $M = 0.01$, an angle of attack of $\alpha = 5$ degrees, and the ambient temperature is 518.7°R. The airfoil has a chord length of $c = 1$ foot. The lift

coefficient is obtained from pressure and skin-friction integration on the airfoil surface in the CFD solution, which is $c_l = 0.9387$.

Given: Coordinates of the points making up the region being considered, and the velocity data $\vec{V} = (u, v)$, density data, and pressure data at those points.
Solve: Use Eqs. 1-5 to find the lift coefficient of the airfoil from the Kutta-Joukowski Theorem.

With the data provided, and using MATLAB, undertake the following tasks:

1. Use the given freestream conditions to determine the dimensional freestream conditions (Hint: Sutherland's Law can provide the dynamic viscosity $\mu = \mu_{ref}\left(\frac{T}{T_{ref}}\right)^{\frac{3}{2}}\left(\frac{T_{ref}+198.72°R}{T+198.72°R}\right)$ where $\mu_{ref} = 3.62 \times 10^{-7}\frac{lb_f*s}{ft^2}$ and $T_{ref} = 518.7°R$).
2. Take the raw MATLAB data file provided, which is ordered along the direction indicated in Fig. 2, and calculate the contribution to the aerodynamic circulation, $\Gamma_{aerodynamic}$, along the outermost column index using the raw CFD data (Hints: You need to use the trapezoidal rule in order to compute the integral in Eq. (2), and the points are not uniformly spaced)
    a. Use the "quiver" function to generate a figure with the velocity vector along the outermost column index.
    b. Plot the circulation of each element as a function of x.
    c. Present a table with the calculated total circulation and lift coefficient
3. Comment on the workflow of steps 1 and 2 and notable observations related to the circulation theory of lift.

3. Data Structure:

A subset of a CFD simulation was extracted to provide the geometric and flow data for the assignment. Fig. 1 shows a visualization of the subset region. The CFD data is provided as a MATLAB data file, which you can load into MATLAB with the 'Project1_data.mat' command. When loaded, two structures should show up in your workspace. Double-click on the structures to see the variables within the structure. To access each of the variables, use "structure.variable", such as "flow.u" to access the u-component of the velocity for all points in the region.

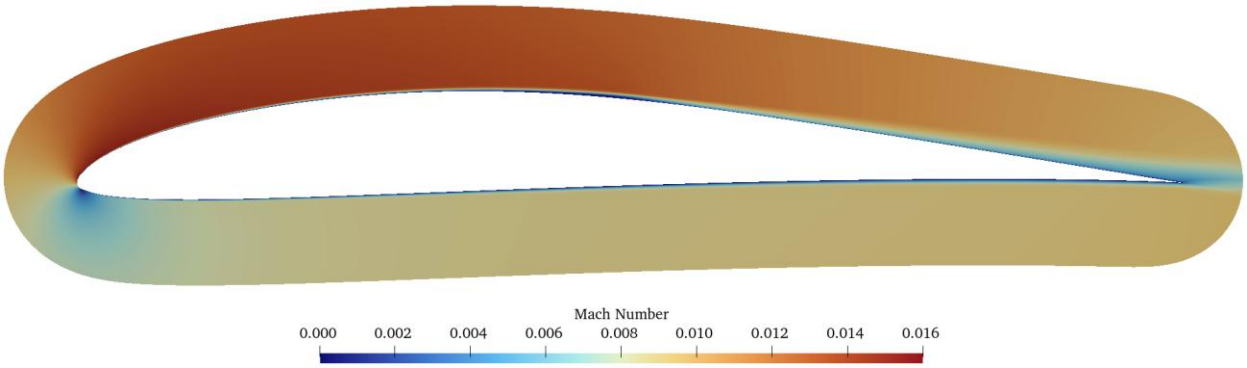


Figure 1: Distribution of Mach number in the assigned region

The CFD grid points form what is known as an "O-grid" with the grid having two directions: around the airfoil and away from the airfoil. These translate into MATLAB as the row indices (501 points) and column indices (121 points). For the primary part of the assignment, you will only need the farthest-out column index.

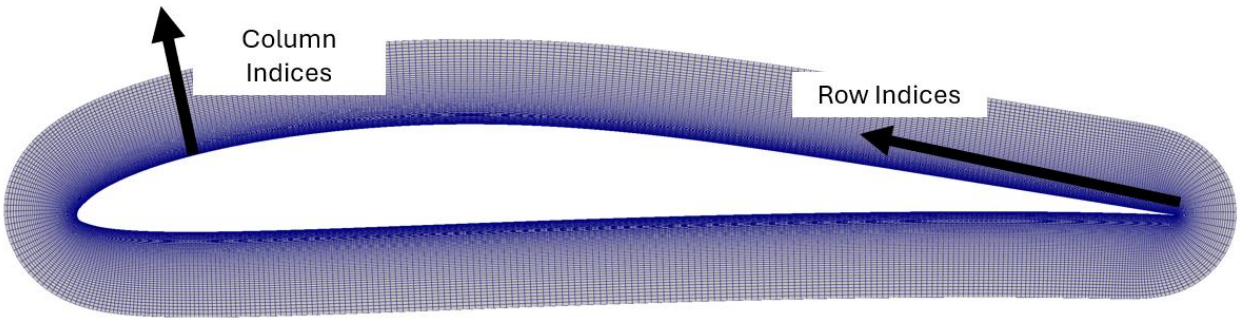


Figure 2: Grid in the assigned region and data indices direction

4. What to submit

Upload a PDF report that contains:

- The name of the student(s) submitting the assignment
- Determined freestream values for part 1 (U_∞ & ρ_∞)
- Figure showing the airfoil and the velocity vectors along the outermost boundary of the provided grid
- Plot of the local circulation contribution vs x
  - Label both axes, **including units**
- The computed aerodynamic circulation and lift coefficient presented in a table
  - Table for circulation value and lift coefficient
  - **Include units for dimensional quantities**
- Short responses to tasks 2 and 3
  - Comment on the results, noteworthy features, comparison with the CFD lift coefficient, etc.
- Include an appendix with your code
- The report should be 2-4 pages, not including the appended code

The assignment is to be uploaded to Canvas by **midnight on 10/16/2026**. A **penalty of 25%** will be applied to late homework. No uploaded work will be accepted after **midnight on 10/16/2026**.

Extra Credit (+15%)

1. Background

Circulation can be calculated in a number of ways, including through a surface integral of the vorticity within the closed contour,

$$\Gamma_{aerodynamic} = -\oiint (\nabla \times \vec{V}) \cdot d\vec{S} \tag{6}$$

Where the curl of the velocity is the vorticity.

2. Task Description

Using Eq. 6, determine the circulation and lift coefficient and compare your result with the calculation from Eq. 2.

3. Tasks

1. Using the raw CFD results, calculate the vorticity throughout the region and present it as a contour plot (similar to the Fig. 1 with the Mach number).
2. Sum the product of the vorticity and area of the cells to determine the total circulation of the region.

4. What to submit

Append to the main PDF report:

- Include the circulation and lift coefficient calculated from the summed vorticity in your table from the main assignment.
    - **Include units for dimensional quantities**
- Using the Matlab "contourf" function, visualize the vorticity throughout the region.
    - Limit the colorbar from -2000 to 2000 without cutting off the values above and below those vorticity values.
    - Provide commentary on the figure, such as where the vorticity is high and low.
- Include an appendix with your code.

5. Hints

- Every cell is a quadrilateral, which can be useful for calculating the area.
- Determining the gradients of the data requires the gradient in both directions, which requires taking derivative with respect to the indices and then using the chain rule to determine the flow gradients (i is row index and j is column index):

$$\frac{\partial u}{\partial i} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial i} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial i}; \frac{\partial u}{\partial j} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial j} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial j}$$

$$\frac{\partial v}{\partial i} = \frac{\partial v}{\partial x}\frac{\partial x}{\partial i} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial i}; \frac{\partial v}{\partial j} = \frac{\partial v}{\partial x}\frac{\partial x}{\partial j} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial j}$$

This can be rearranged as,

$$\begin{bmatrix} \frac{\partial u}{\partial i} \\ \frac{\partial u}{\partial j} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial i} & \frac{\partial z}{\partial i} \\ \frac{\partial x}{\partial j} & \frac{\partial z}{\partial j} \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial v}{\partial i} \\ \frac{\partial v}{\partial j} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial i} & \frac{\partial z}{\partial i} \\ \frac{\partial x}{\partial j} & \frac{\partial z}{\partial j} \end{bmatrix} \begin{bmatrix} \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial y} \end{bmatrix}$$

to be solved for the velocity gradients with respect to space. The other gradients can be determined with the "gradient" function.

- It is easier to determine the vorticity at the points where you have the other flow data and then calculate the value for a cell by averaging its corners.